

1. Administrative & Study Identification

Full Study Title: A Pharmacokinetics (PK), Pharmacodynamics (PD), Safety and Tolerability Study of Fenebrutinib in Children and Adolescents With Relapsing Multiple Sclerosis (RMS) **Short Title/Acronym:** CN45847 (Fenebrutinib in Children/Adolescents with RMS) **Protocol Number:** CN45847 (Secondary ID: 2024-519800-28-00) **NCT Number:** NCT07161258 **Sponsor Name:** Hoffmann-La Roche **Investigational Product:** Fenebrutinib (Bruton's tyrosine kinase [BTK] inhibitor) **Phase of Development:** Phase II **Medical Monitor:** Hoffmann-La Roche Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective **Primary Objective:** To evaluate the pharmacokinetics (PK), pharmacodynamic (PD) effects, safety, and tolerability of fenebrutinib in children and adolescents with relapsing multiple sclerosis (RMS). **Secondary Objectives:** To evaluate the clinical efficacy of fenebrutinib in preventing signs of RMS (e.g., MS relapses, new brain lesions on MRI, and disability worsening) during an optional extension period.

2.2 Background & Rationale Relapsing Multiple Sclerosis (RMS) is a condition where the immune system attacks the protective lining of nerve fibers in the brain and spinal cord. There is a critical need for effective, safe, and easily administered treatments to prevent disease worsening in pediatric populations. Fenebrutinib is a selective, oral Bruton's tyrosine kinase (BTK) inhibitor that offers a highly targeted approach to reduce neuroinflammation, potentially causing fewer side effects compared to broad-acting therapies. This study aims to characterize the drug's safety profile and determine the appropriate dose for children and teenagers (aged 10 to <18 years) for application in future clinical trials.

3. Study Design & Methodology

3.1 Design Framework **Study Type:** Interventional **Control Type:** Single-arm **Blinding:** Open-label **Randomization:** N/A

3.2 Planned Enrollment & Duration **Target N:** To be determined (Estimates not publicly disclosed for this Phase 2 PK/PD stage) **Number of Sites:** Multicenter (Global) **Estimated Study Start:** 2024/2025 **Estimated Primary Completion:** TBD

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 10 to < 18 years (10 to 17 years old). 2. Diagnosis of RMS in accordance with the International Pediatric Multiple Sclerosis Study Group (IPMSSG) criteria (Version 2012) and the revised 2017 McDonald Criteria. 3. Expanded Disability Status Scale (EDSS) at screening from 0 to 5.5 points, inclusive. 4. At least one MS relapse during the previous year, OR two MS relapses in the previous 2 years, OR evidence of at least one Gd-enhancing lesion on MRI within 6 months. 5. Have received all childhood vaccinations as per local/national recommendations.

4.2 Exclusion Criteria 1. Diagnosis of primary progressive multiple sclerosis (PPMS) or non-active secondary progressive multiple sclerosis (SPMS). 2. Recent or anticipated use of prohibited medications/treatments, including certain disease-modifying therapies (DMTs), immunosuppressants, or strong CYP3A4 inhibitors (drugs interacting with fenebrutinib). 3. Co-morbid conditions such as potentially confounding neurological, somatic, or metabolic disorders; active infections; or a history of cancer, transplants, or bleeding disorders. 4. Inability to complete an MRI scan or receive gadolinium contrast. 5. Sensitivity or intolerance to any ingredient of fenebrutinib tablets, or abnormal liver function tests/blood counts.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Fenebrutinib administered as daily oral tablets. **Route of Administration:** Oral. **Duration of Treatment:** A 3-month Dose Exploration Period, followed by an Optional Extension Period lasting approximately 2 years.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care for managing MS symptoms. **Prohibited:** Certain DMTs, other immunosuppressants, anticoagulants, strong CYP3A4 inhibitors, and certain live vaccines.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
1	1 Month Prior	Informed Consent/Assent, Medical History, EDSS scoring, MRI, Psychiatric risk screening (C-SSRS/PHQ-A), Liver/Blood tests.
2	Day 0	First Dose, PK/PD Sampling Initialization.
3	Month 3	End of Dose Exploration Period; Comprehensive Safety Labs; decision on entering the Optional Extension Period.
4	~Year 2	Final clinical EDSS/MRI Assessment, Follow-up for extended safety.

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Characterization of Pharmacokinetics (PK) and Pharmacodynamics (PD) parameters of fenebrutinib, along with safety/tolerability measures (incidence and severity of Adverse Events). **Measurement Method:** Serial venous blood sampling and continuous clinical safety evaluations.

7.2 Secondary & Exploratory Endpoints 1. Efficacy metrics: Annualized Relapse Rate (ARR) and prevention of MS relapses. 2. Radiological metrics: Number of new or enlarging Gd-enhancing brain lesions on MRI. 3. Clinical metrics: Changes in Expanded Disability Status Scale (EDSS) over time.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard clinical monitoring including liver function tests, blood counts, and close tracking of psychiatric/suicidal behavior using the C-SSRS and PHQ-A scales. **Serious Adverse Events (SAEs):** Evaluated continuously and reported per standardized trial guidelines (typically 24 hours). **Data Monitoring Committee (DMC):** Internal and external safety oversight managed by Hoffmann-La Roche.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: PK/PD analysis set (for dose determination) and Safety population. **Sample Size Justification:** Scaled appropriately to determine dose exposure, PK, PD, and basic tolerability parameters in a pediatric/adolescent subgroup. **Handling of Missing Data:** Standard imputation rules applied per sponsor SAP.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** High-frequency onsite and remote monitoring expected for Phase II pediatric trials. **Audit Trail:** Validated eCRF tracking and source data verification logs.

11. Study Identifiers & Governance

Primary ID: CN45847 **Registry Number:** NCT07161258 **Sponsor/Lead Organization:** Hoffmann-La Roche **Study Title:** A Pharmacokinetics (PK), Pharmacodynamics (PD), Safety and Tolerability Study of Fenebrutinib in Children and Adolescents With Relapsing Multiple Sclerosis (RMS) **Official Regulatory Title:** An Open-label, Single-

arm Study to Evaluate Pharmacokinetics, Pharmacodynamic Effects, Safety and Tolerability of Fenebrutinib in Children and Adolescents With Relapsing Multiple Sclerosis

12. Clinical Framework (Multiple Sclerosis Specific)

Primary Condition: Relapsing Multiple Sclerosis (RMS) in Children and Adolescents **Diagnosis Threshold:** 2017 McDonald Criteria / 2012 IPMSSG criteria (EDSS 0 to 5.5) **Medical Methodology:** Disease-Modifying Therapy (DMT) / Bruton's Tyrosine Kinase (BTK) Inhibition **Optimization Target:** Pediatric dose determination based on PK/PD and Safety

13. Study Procedures & Methodology

Management Pathway: Daily oral fenebrutinib administration and routine MS clinical assessments **Education Protocol (HCP):** Protocol training, pediatric MS assessment, and psychiatric risk monitoring **Education Protocol (Patient):** Informed assent/consent and study requirement adherence (e.g., tablet swallowing, ability to complete manual dexterity/walking tests and MRI scans) **Quality Audit Mechanism:** Source data verification, detailed laboratory tracking, and AE/SAE reporting

14. Observation & Follow-Up Schedule

Total Duration: Up to ~27 months (3-month Dose Exploration + ~24-month Optional Extension) **Onsite Visit Cadence:** Screening, Baseline, 3-Month interim, and periodic maintenance visits during the extension phase **Remote Monitoring Frequency:** As dictated by the site's standard protocol for AE follow-up **Checklist Review Interval:** Ongoing AE, concomitant medication, and psychiatric well-being review at every evaluation point

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]			Lead	
Statistician	[Name]	_____	[DD-MMM-YYYY]				